United States Patent	12414287
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Lee; Yuna et al.

Method of fabricating semiconductor device

Abstract

A method of fabricating a semiconductor device includes forming an insulating layer and a peripheral structure on first and second regions of the substrate, forming first and second mask layers on the insulating layer and the peripheral structure, patterning the first and second mask layers to form first and second mask structures on the first and second regions, etching the insulating layer using the first and second mask structures as an etching mask, to form insulating patterns, forming a sacrificial layer in spaces between two adjacent insulating patterns on the first region, removing the second mask pattern on the first region by a dry etching process, forming an anti-oxidation layer on a surface of the second mask layer on the second region after removing the second mask pattern on the first region, and removing the second mask layer with the anti-oxidation layer by a wet etching process.

Inventors: **Lee; Yuna (Hwaseong-si, KR), Park; Sangwuk (Hwaseong-si, KR), Yoon; Hyunchul (Seongnam-si, KR), Lee; Seungjae (Seoul, KR), Rhee; Joonkyu (Hwaseong-si, KR), Lee; Chanmin (Hwaseong-si, KR), Hong; Jungpyo (Gwangmyeong-si, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

Family ID: **86896671**

Assignee: **SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)**

Appl. No.: **18/088370**

Filed: **December 23, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230209802 A1	Jun. 29, 2023

Foreign Application Priority Data

KR	10-2021-0189553	Dec. 28, 2021

Publication Classification

Int. Cl.: **H10B12/00** (20230101)

U.S. Cl.:

CPC **H10B12/09** (20230201); **H10B12/482** (20230201); **H10B12/50** (20230201);

Field of Classification Search

CPC: H10B (12/09); H10B (12/482); H10B (12/50)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7550363	12/2008	Lee	438/424	H01L 21/76229
8551877	12/2012	Ranjan et al.	N/A	N/A
9653321	12/2016	Mikami	N/A	N/A
10186597	12/2018	Ahn et al.	N/A	N/A
11011521	12/2020	Korkmaz et al.	N/A	N/A
2009/0023277	12/2008	Kang	257/E21.177	H10D 64/027
2014/0264672	12/2013	Park et al.	N/A	N/A
2018/0166529	12/2017	Park et al.	N/A	N/A
2019/0103302	12/2018	Yoon	N/A	H01L 21/76895
2019/0348418	12/2018	Hwang et al.	N/A	N/A
2020/0312852	12/2019	Ryu et al.	N/A	N/A
2021/0057339	12/2020	Lee et al.	N/A	N/A
2021/0066303	12/2020	Huang	N/A	N/A
2021/0125998	12/2020	Kim et al.	N/A	N/A
2021/0151439	12/2020	Choi et al.	N/A	N/A
2021/0272961	12/2020	Tung et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1020130102505	12/2012	KR	N/A
1020150043978	12/2014	KR	N/A
202109835	12/2020	TW	N/A

Primary Examiner: Booth; Richard A

Attorney, Agent or Firm: Muir Patent Law, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims benefit of priority to Korean Patent Application No. 10-2021-0189553 filed on Dec. 28, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

(2) The present disclosure relates to a method of fabricating a semiconductor device, including a method of patterning a material layer using at least two mask layers, and a semiconductor device fabricated by the same.

(3) Research on reductions of sizes of elements, constituting a semiconductor device, and improvements in performance thereof has been conducted. For example, research on reliable and stable formation of elements, having reduced sizes, in a dynamic random access memory (DRAM) has been conducted.

SUMMARY

(4) Example embodiments provide a method of fabricating a semiconductor device, including a method of stably patterning a material layer using at least two mask layers.

(5) Example embodiments provide a semiconductor device fabricated by the above-described method.

(6) According to an example embodiment, a method of fabricating a semiconductor device includes forming a plurality of bitline structures on a first region of a substrate, forming a peripheral device structure on a second region, adjacent to the first region, of the substrate, forming an insulating layer in spaces between two adjacent bitline structures of the plurality of bitline structures, forming a first mask layer and a second mask layer sequentially on the insulating layer, the bitline structures, and the peripheral device structure, patterning the first and second mask layers to form a first mask structure including a first mask pattern and a second mask pattern sequentially stacked on the first region, and a second mask structure including the first and second mask layers remaining, after the patterning of the first and second mask layers, on the second region, etching the insulating layer by an etching process using the first and second mask structures as an etching mask, to form a plurality of insulating patterns in spaces between two adjacent bitline structures of the plurality of bitline structures, forming a sacrificial layer to fill spaces between two adjacent insulating patterns of the plurality of insulating patterns on the first region, removing the second mask pattern on the first region by performing a dry etching process, forming an anti-oxidation layer on a surface of the second mask layer on the second region after removing the second mask pattern on the first region, removing the second mask layer, having a surface on which the anti-oxidation layer is formed, by performing a wet etching process, removing the sacrificial layer to form a plurality of fence holes after selectively removing the second mask layer on which the anti-oxidation layer is formed, forming a plurality of insulating fences in the plurality of fence holes, respectively, removing the first mask pattern and the plurality of insulating patterns to form a plurality of contact holes, and forming a plurality of contact plugs in the plurality of contact holes, respectively.

(7) According to an example embodiment, a method of fabricating a semiconductor device includes forming an insulating layer on a first region of a substrate and a peripheral structure on a second region of the substrate, forming a first mask layer and a second mask layer sequentially on the insulating layer and the peripheral structure, patterning the first and second mask layers to form a first mask structure including a first mask pattern and a second mask pattern sequentially stacked on the first region, and a second mask structure including the first and second mask layers remaining, after the patterning of the first and second mask layers, on the second region, etching

the insulating layer by an etching process using the first and second mask structures as an etching mask, to form a plurality of insulating patterns separated apart from each other, forming a sacrificial layer in spaces between two adjacent insulating patterns of the plurality of insulating patterns, on the first region, removing the second mask pattern on the first region by performing a dry etching process, forming an anti-oxidation layer on a surface of the second mask layer on the second region after removing the second mask pattern on the first region, and removing the second mask layer, having a surface on which the anti-oxidation layer is formed, by performing a wet etching process.
(8) According to an example embodiment, a method of fabricating a semiconductor device includes forming a cell transistor at a first region of a substrate, forming a plurality of bitline structures and a peripheral device structure on the substrate, the plurality of bitline structures being formed on the first region and the peripheral device structure being formed on a second region, adjacent to the first region, of the substrate, forming an insulating layer between spaces between two adjacent bitline structures of the plurality of bitline structures, forming a first mask layer and a second mask layer sequentially on the insulating layer, the plurality of bitline structures, and the peripheral device structure, patterning the first and second mask layers to form a first mask structure including a first mask pattern and a second mask pattern sequentially stacked on the first region, and a second mask structure including the first and second mask layers remaining, after the patterning of the first and second layers, on the second region, etching the insulating layer by an etching process, using the first and second mask structures as an etching mask, to form a plurality of insulating patterns, each insulating pattern of the plurality of insulating patterns being disposed between corresponding two adjacent bitline structures of the plurality of bitline structures, forming a sacrificial layer to fill spaces between two adjacent insulating patterns of the plurality of insulating patterns, on the first region, removing the second mask pattern by performing a dry etching process, forming an anti-oxidation layer on a surface of the second mask layer on the second region after removing the second mask pattern, and removing the second mask layer, having a surface on which the anti-oxidation layer is formed, by performing a wet etching process.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings.
(2) FIGS. **1**A and **1**B are process flow diagrams illustrating an example of a method of fabricating a semiconductor device according to an example embodiment.
(3) FIG. **2** is a plan view illustrating an example of a method of fabricating a semiconductor device according to an example embodiment.
(4) FIGS. **3**A, **3**B, **4**A, **4**B, **5**A, **5**B, **6**A, **6**B, **7**A, **7**B, **8**A, **8**B, **9**A, **9**B, **10**, **11**A, **11**B, **12**A, **12**B, **13**A and **13**B are cross-sectional views illustrating a method of fabricating a semiconductor device according to an example embodiment.
(5) FIG. **14** is a process flowchart illustrating another example of a method of fabricating a semiconductor device according to an example embodiment.
(6) FIGS. **15**A, **15**B, **16**A, **16**B, **17**A, **17**B, **18**A, **18**B, **19**, and **20** are cross-sectional views illustrating another example of a method of fabricating a semiconductor device according to an example embodiment.
(7) FIGS. **21** and **22** are cross-sectional views illustrating an example of a method of fabricating a semiconductor device according to an example embodiment.
DETAILED DESCRIPTION

(8) Hereinafter, terms such as “upper,” “intermediate,” and “lower” may be replaced with other terms, for example, “first,” “second,” and “third” to describe components of the present specification. The terms such as “first,” “second,” and “third” may be used to describe various components, but the components may not be restricted by the terms, and “first component” may be referred to as “second component.”
(9) Hereinafter, methods of fabricating a semiconductor device according to example embodiments and structures of a semiconductor device, fabricated by the methods, will be described.
(10) A method of fabricating a semiconductor device according to an example embodiment and a structure of a semiconductor device fabricated by the method will be described with reference to FIGS. **1**A to **13**B. FIGS. **1**A and **1**B are process flow diagrams illustrating an example of a method of fabricating a semiconductor device according to an example embodiment, FIG. **2** is a plan view illustrating an example of a method of fabricating a semiconductor device according to an example embodiment, and FIGS. **3**A to **13**B are cross-sectional views illustrating a method of fabricating a semiconductor device according to an example embodiment. In FIGS. **3**A to **13**B, FIGS. **3**A, **4**A, **5**A, **6**A, **7**A, **8**A, **9**A, **11**A, **12**A, and **13**A are cross-sectional views illustrating regions taken along lines I-I′ and II-II′ of FIG. **2**, and FIGS. **3**B, **4**B, **5**B, **6**B, **7**B, **8**B, **9**B, **10**, **11**B, **12**B, and **13**B are cross-sectional views illustrating regions taken along lines and IV-IV′ of FIG. **2**.
(11) Referring to FIGS. **1**A, **2**, **3**A, and **3**B, in operation S**5**, a lower structure LS may be formed. The forming of the lower structure LS may include forming a device isolation layer **6***s* on a semiconductor substrate **3** to define cell active regions **6***a***1** and a peripheral active region **6***a***2**, forming gate trenches **12** to intersect the cell active region **6***a***1** and to extend inwardly of the device isolation layer **6***s*, and forming cell gate structures GS**1** to respectively fill the gate trenches **12** and cell gate capping layers **18** on the cell gate structures GS**1**.
(12) The semiconductor substrate **3** may be formed of a semiconductor material such as silicon.
(13) Each of the cell gate structures GS**1** may include a cell gate dielectric layer **14**, conformally covering an internal wall of the gate trench **12**, and a cell gate electrode **16** filling a portion of the gate trench **12** on the cell gate dielectric layer **14**.
(14) The forming of the lower structure LS may further include forming a gate capping layer **18** to fill a remaining portion of the gate trench **12** on the gate electrode **16**.
(15) The gate electrode **16** may include or may be formed of doped polysilicon, metal, a conductive metal nitride, a metal-semiconductor compound, a conductive metal oxide, graphene, carbon nanotubes, or combinations thereof. For example, the gate electrode **16** may be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrO.sub.x, RuO.sub.x, graphene, carbon nanotubes, or combinations thereof, but example embodiments are not limited thereto. The gate electrode **16** may be a single layer or multiple layers of the above-mentioned materials. For example, the gate electrode **16** may include a first electrode layer, which may be formed of a metal material, and a second electrode layer which may be formed of doped polysilicon on the first electrode layer. The gate capping layer **18** may include or may be formed of an insulating material, for example, silicon nitride.
(16) The forming of the lower structure LS may further include forming cell sources/drains SD**1**, including a first impurity region **9***a* and a second impurity region **9***b*, at the cell active regions **6***a***1** using an ion implantation process.
(17) The cell gate structure GS**1** and the cell sources/drains SD**1** may constitute cell transistors TR**1**.
(18) In an embodiment, the cell source/drain regions SD**1** may be formed before the device isolation layer **6***s* is formed.
(19) In an embodiment, the cell source/drain regions SD**1** may be formed after the device isolation layer **6***s* is formed and before the gate trenches **12** are formed.
(20) In an embodiment, the cell source/drain regions SD**1** may be formed after the cell gate

structures GS**1** and the cell gate capping layer **18** are formed.
(21) The cell active regions **6***a***1** may be formed of single-crystalline silicon. The cell active regions **6***a***1** may have P-type conductivity, and the first and second impurity regions **9***a* and **9***b* may have N-type conductivity.
(22) The lower structure LS may be formed in a first region MA and a second region PA. When the semiconductor device **1** according to example embodiments is a memory device (for example, a DRAM device), the first region MA may be a memory cell array region and the second region PA may be a peripheral circuit region around the memory cell array region.
(23) In example embodiments, the first region MA may be referred to as a memory cell array region or a memory region, and the second region PA may be referred to as a peripheral circuit region or a peripheral region.
(24) Referring to FIGS. **1**A, **2**, **4**A, and **4**B, a buffer insulating layer **21** may be formed on the lower structure LS in the first region MA. In an embodiment, the buffer insulating layer **21** may not be formed on the lower structure LS in the second region PA. The buffer insulating layer **21** may include at least a silicon oxide layer and a silicon nitride layer sequentially stacked.
(25) In operation S**10**, interconnection structures BS and peripheral device structures TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129** may be formed. The interconnection structures BS and the peripheral device structures TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129** may be formed on the lower structure LS. A portion of the interconnection structures BS and a portion of the peripheral device structures TR**2**, **130**, **128***a*, **128***b*, **128***c* and **129** may be simultaneously formed.
(26) The peripheral device structures TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129** may be referred to as peripheral structures.
(27) A line-shaped opening **33** may be formed between the interconnection structures BS.
(28) The interconnection structures BS may be formed in the first region MA, and the peripheral device structures TR**2**, **130**, **128***a*, **128***b* and **128***c* may be formed in the second region PA.
(29) When viewed in a plan view, each of the gate structures GS**1** may extend in a first direction X, and each of the interconnection structures BS may extend in a second direction Y, perpendicular to the first direction X.
(30) The forming of each of the interconnection structures BS may include forming a conductive line **25** and an interconnection capping layer **28** sequentially stacked and forming insulating spacers **30** and **31** on side surfaces of the conductive line **25** and side surfaces of the interconnection capping layer **28**.
(31) In each of the interconnection structures BS, the conductive line **25** may include a first layer **25***a*, a second layer **25***b*, and a third layer **25***c* sequentially stacked, and a portion of the first layer **25***a* may extend downwardly to form a plug portion **25***p*, electrically connected to the first impurity region **9***a*, of the first source/drain regions SD**1**. In an embodiment, the plug portion **25***p* may contact the first impurity region **9***a*. It will be understood that when an element is referred to as being “connected” or “coupled” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element, there are no intervening elements present at the point of contact. As used herein, components described as being “electrically connected” are configured such that an electrical signal can be transferred from one component to the other (although such electrical signal may be attenuated in strength as it transferred and may be selectively transferred).
(32) In the conductive line **25**, the first layer **25***a* may be formed of a doped silicon layer, and the second layer **25***b* may be formed of a metal-semiconductor compound layer (for example, WN, TiN, or the like), and the third layer **25***c* may be formed of a metal layer (for example, W, or the like).
(33) In an example embodiment, the interconnection structures BS may be bitline structures. For

example, the conductive line **25** may be a bitline including a plug portion **25***p* electrically connected to the first impurity region **9***a*. The conductive line **25** may be a bitline of a memory device such as a DRAM.

(34) The insulating spacers **30** and **31** may include a first spacer portion **30** and a second spacer portion **31**. The first spacer portion **30** may cover a side surface of the plug portion **25***p*. The second spacer portion **31** may cover side surfaces of the conductive line **25** and surfaces of the interconnection capping layer **28**. The second spacer portion **31** may be positioned at a level higher than a level of the plug portion **25***p*. The present invention is not limited thereto. In an embodiment, the first spacer portion **30** and the second spacer portion **31** may be integrally formed in the same fabrication process.

(35) The interconnection capping layer **28** may include a first layer **28***a*, a second layer **28***b*, and a third layer **28***c* sequentially stacked. The interconnection capping layer **28** may be formed of a silicon nitride and/or a silicon nitride-based insulating material.

(36) The peripheral device structures TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129** may include a peripheral transistor TR**2**.

(37) The peripheral transistor TR**2** may include second peripheral sources/drains SD**2**, spaced apart from each other in the peripheral active region **6***a***2**, and a peripheral gate GS**2** formed on a peripheral active region between the second peripheral sources/drains SD**2**.

(38) The peripheral gate GS**2** may include a peripheral gate dielectric **123** and a peripheral gate electrode **125** disposed on the peripheral gate dielectric **123**.

(39) The peripheral gate electrode **125** may include a first layer **125***a*, a second layer **125***b*, and a third layer **125***c*, sequentially stacked.

(40) At least a portion of the peripheral gate electrode **125** may be formed of substantially the same material as at least a portion of the conductive line **25**. For example, the first layer **125***a* of the peripheral gate electrode **125** and the first layer **25***a* of the conductive line **25** may be formed of a doped silicon layer, the second layer **125***b* of the peripheral gate electrode **125** and the second layer **25***b* of the conductive line **25** may be formed of a metal-semiconductor compound layer (for example, WN, TiN, or the like), and the third layer **125***c* of the peripheral gate electrode **125** and the third layer **25***c* of the conductive line **25** may be formed of a metal layer (for example, W, or the like). In an embodiment, the second layer **125***b* of the peripheral gate electrode **125** and the second layer **25***b* of the conductive line **25** may be formed of the same metal-semiconductor compound layer in the same fabrication process. In an embodiment, the third layer **125***c* of the peripheral gate electrode **125** and the third layer **25***c* of the conductive line **25** may be formed of the same metal layer in the same fabrication process.

(41) The peripheral device structure TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129** may further include a peripheral capping layer **128***a*, formed on the peripheral gate GS**2**, and peripheral spacers **129** formed on side surfaces of the peripheral gate GS**2** and side surfaces the peripheral capping layer **128***a*.

(42) The peripheral capping layer **128***a* may be formed of a silicon nitride.

(43) The peripheral spacers **129** may include at least one of silicon oxide, silicon oxynitride, and silicon nitride.

(44) The peripheral device structure TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129** may further include an insulating liner **128***b* conformally covering the peripheral transistor TR**2**, the peripheral capping layer **128***a*, and the peripheral spacers **129**, an interlayer insulating layer **130** on the insulating liner **128***b*, and an upper insulating layer **128***c* on the interlayer insulating layer **130**. The interlayer insulating layer **130** may be formed on the insulating liner **128***b* on side surfaces of the peripheral gate GS**2** and the peripheral capping layer **128***a*. The insulating liner **128***b* may be formed of silicon nitride or a silicon nitride-based material. The interlayer insulating layer **130** may be formed of silicon oxide. The upper insulating layer **128***c* may be formed of silicon nitride or a silicon nitride-based material.

(45) Referring to FIGS. **1**A, **2**, **5**A, and **5**B, in operation S**15**, an insulating layer **135** may be formed between the interconnection structures BS. The insulating layer **135** may include or may be formed of silicon oxide, but example embodiments are not limited thereto. For example, the insulating layer **135** may be a material which may fill the opening **33** between the interconnection structures BS without a void. In some embodiments, the insulating layer **135** may be formed in spaces between two adjacent interconnection structures of the interconnection structures BS, completely filling the spaces.
(46) In operation S**20**, a first mask layer **138** and a second mask layer **141** may be sequentially formed. The first and second mask layers **138** and **141**, sequentially stacked, may be formed on the interconnection structures BS, the insulating layer **135**, and the upper insulating layer **128***c*.
(47) The first and second mask layers **138** and **141** may be formed of materials having etch selectivity with respect to the material of the upper insulating layer **128***c* and the interconnection capping layer **28**. The first and second mask layers **138** and **141** may be formed of materials having different levels of etch selectivity with respect to the material of the upper insulating layer **128***c* and the interconnection capping layer **28**. For example, the upper insulating layer **128***c* and the interconnection capping layer **28** may be formed of silicon nitride and/or a silicon nitride-based insulating material, the first mask layer **138** may be formed of oxide (e.g., silicon oxide) or an oxide-based material, and the second mask layer **141** may be formed of polysilicon. For example, the first mask layer **138** may be formed of a silicon oxide.
(48) Referring to FIGS. **1**A, **2**, **6**A and **6**B, in operation S**25**, the first and second mask layers (**138** and **141** of FIGS. **5**A and **5**B) in the first region MA may be patterned to form a first mask structure **143***a*, including a first mask pattern **138***a* and a second mask pattern **141***a* sequentially stacked in the first region MA, and a second mask structure **143***b* including the first and second mask layers **138***b* and **141***b* remaining in the second region PA.
(49) In operation S**30**, the insulating layer **135** of FIGS. **5**A and **5**B may be etched by an etching process using the first and second mask structures **143***a* and **143***b* as etching masks to form insulating patterns **135***a* between the interconnection structures BS. While forming the insulating patterns **135***a*, a portion of a lower portion of the insulating layer (**135** of FIGS. **5**A and **5**B) may be etched. For example, openings **145** may be formed between the insulating patterns **135***a*, and the openings **145** may extend downwardly to penetrate through the buffer insulating layer **21**.
(50) In some embodiments, the openings **145** may be referred to as fence holes.
(51) During a time when the etching process of forming the insulating patterns **135***a* is performed, a portion of the second mask pattern **141***a* of the first mask structure **143***a* may be etched, and a portion of the mask layer **141***b* of the second mask structure **143***b* may be etched.
(52) In an example embodiment, a thickness of the second mask layer **141***b* of the second mask structure **143***b* may be greater than a thickness of the second mask pattern **141***a* of the first mask structure **143***a*.
(53) Referring to FIGS. **1**A, **2**, **7**A, and **7**B, in operation S**40**, a sacrificial layer **148** may be formed to fill at least a space between the insulating patterns **135***a*.
(54) The sacrificial layer **148** may cover a portion of a side surface of the first mask pattern **138***a* while filling the space between the insulating patterns **135***a*. For example, an upper surface of the sacrificial layer **148** may be formed at a level, higher than or equal to a level of a lower surface of the first mask pattern **138***a* and lower than a level of the second mask pattern **141***a*. The sacrificial layer **148** may be formed of a spin-on-hardmask (SOH) material.
(55) After the sacrificial layer **148** is formed, the second mask layer **141***b* of the second mask structure **143***b* may have a first thickness T**1**.
(56) Referring to FIGS. **1**B, **2**, **8**A, and **8**B, in operation S**50**, the structure formed up to the sacrificial layer **148** may be loaded into a process chamber. The process chamber may be a plasma process chamber. In some embodiments, the semiconductor substrate **3** with the sacrificial layer **148** may be loaded into a process chamber immediately after the formation of the sacrificial layer

148.
(57) In operation S**55**, in the process chamber, the second mask pattern **141***a* may be removed by a dry etching process. The dry etching process may be a plasma etching process.
(58) The removing the second mask pattern **141***a* may include performing a first etching process of removing a native oxide, which may be formed on a surface of the second mask pattern **141***a*, and performing a second etching process of removing the second mask pattern **141***a*. When the second mask pattern **141***a* is formed of polysilicon, the second mask pattern **141***a* may be removed by a plasma etching process using a Cl.sub.2-based source gas.
(59) A thickness of the second mask layer **141***b* of the second mask structure **143***b* of FIG. **7**B may be reduced during a time when the dry etching process of removing the second mask pattern **141***a* is performed. Accordingly, the second mask layer (**141***b* of FIG. **7**B) may be formed as a second mask layer **141***b*′ having a second thickness T**2**, smaller than the first thickness (T**1** of FIG. **7**B). Thus, the second mask structure (**143***b* of FIG. **7**B) may be formed as a second mask structure **143***b*′ including the second mask layer **141***b*′ having a reduced thickness compared to the thickness of the second mask layer **141***b* of FIGS. **7**A and **7**B.
(60) Referring to FIGS. **1**B, **2**, **9**A, and **9**B, an anti-oxidation layer **141***p* may be formed on a surface of the second mask layer **141***b*′ in the second region PA. In operation S**60**, for example, in the process chamber, the anti-oxidation layer **141***p* may be formed on the surface of the second mask layer **141***b*′ in the second region PA by performing a hydrogen plasma treatment process **151**. The anti-oxidation layer **141***p* may be formed by bonding silicon elements of polysilicon of the second mask layer **141***b*′ to hydrogen elements supplied by the hydrogen plasma treatment process **151**. In some embodiments, in the hydrogen plasma treatment process **151**, dangling bonds of polysilicon at the surface of the second mask layer **141***b*′ may be terminated with hydrogen atoms, thereby preventing a native oxide from forming at the surface of the second mask layer **141***b*′. Accordingly, a second mask structure **143***b*″, including the second mask layer **141***b*′ having a surface on which the anti-oxidation layer **141***p* is formed, may be formed.
(61) In operation S**65**, a structure formed up to the anti-oxidation layer **141***p* may be unloaded from the process chamber. In an embodiment, the semiconductor substrate **3** with the anti-oxidation layer **141***p* may be unloaded from the process chamber immediately after the formation of the anti-oxidation layer **141***p*.
(62) Referring to FIGS. **1**B, **2**, and **10**, the second mask layer (**141***b*′ of FIG. **9**B), having the surface on which the anti-oxidation layer (**141***p* of FIG. **9**B) is formed, may be selectively removed. In operation S**70**, for example, the second mask layer (**141***b*′ of FIG. **9**B), having the surface on which the anti-oxidation layer (**141***p* of FIG. **9**B) is formed, may be removed by a wet etching process. For example, the second mask layer (**141***b*′ of FIG. **9**B) formed of polysilicon may be removed by a wet etching process using NH.sub.4OH and H.sub.2O.sub.2 as an etchant. In some embodiments, when the second mask layer **141***b*′ of FIG. **9**B is removed, the anti-oxidation layer which is formed on the second mask layer **141***b*′ is removed.
(63) Referring to FIGS. **1**B, **2**, **11**A, and **11**B, in operation S**75**, the sacrificial layer **148** may be selectively removed to form fence holes **145**. The fence holes **145** may be substantially the same as the openings of FIG. **6**B. During a time when the sacrificial layer **148** is selectively removed, the first mask pattern **138***a* and the first mask layer **138***b* may remain unetched.
(64) Referring to FIGS. **1**B, **2**, **12**A, and **12**B, insulating fences **155** may be formed in the fence holes **145**. The insulating fences **155** may be formed of silicon nitride or a silicon nitride-based insulating material.
(65) In an embodiment, the first mask pattern **138***a* and the first mask layer **138***b* may be removed during formation of the insulating fences **155**.
(66) In an embodiment, the first mask pattern **138***a* and the first mask layer **138***b* may remain after formation of the insulating fences **155**.
(67) Referring to FIGS. **1**B, **2**, **13**A, and **13**B, in operation S**85**, first contact holes **157** may be

formed. In the formation of the first contact holes **157**, an etching process using the insulating fences **155**, the interconnection structures BS, and the upper insulating layer **128***c* as etching masks may be performed such that a portion of lower portions of the insulating patterns (**135***a* of FIGS. **12**A and **12**B) are etched to form the first contact hole **157**. For example, the buffer insulating layer **21** below the insulating patterns **135***a* of FIGS. **12**A and **12**B may be etched. The second impurity regions **9***b* and the first contact holes **157** may overlap each other in a vertical direction Z. In operation S**90**, first contact plugs **60** may be formed in the first contact holes **157**. In some embodiments, the first contact holes **157** may expose the second impurity regions **9***b*. The exposed second impurity regions **9***b* may be recessed during a time when the first contact holes **157** are formed. The first contact plugs **60** may contact the exposed second impurity regions **9***b*.

(68) Second contact plugs **160***s* and **160** may be formed to be electrically connected to the second sources/drains SD**2** in the second region PA while the first contact plugs **60** are formed in the first region MA.

(69) First pads **60***p* on the first contact plugs **60** and second pads **160***p* on the second contact plugs **160***s* and **160** may be formed while forming the first contact plugs **60** and the second contact plugs **160***s* and **160**. In some embodiments, the first pads **60***p* and the first contact plugs **60** may be integrally formed in the same process. In some embodiments, the second pads **160***p* and the plug portions **160** of the second contact plugs **160***s* and **160** may be integrally formed in the same process.

(70) In an example, in the formation of the first and second contact plugs **60**, **160***s* and **160** and the first and second pads **60***p* and **160**P, a preliminary layer may be formed to fill the first contact holes **157**. A second contact hole **158** may be formed to sequentially penetrate through the upper insulating layer **128***c*, the interlayer insulating layer **130**, and the insulating liner **128***b* in the second region PA. The second contact hole **158** may expose the second sources/drains SD**2**. A portion of the preliminary layer may be etched to form a lower layer **60***a* in the first contact holes **157**. A first metal-semiconductor compound layer **60***b* and a second metal-semiconductor compound layer **160***s* may be simultaneously formed on the lower layer **60***a* in the first contact holes **157** and the second sources/drains SD**2** exposed by the second contact hole **158**, respectively. An upper layer **60***c* filling a remaining portion of the first contact holes **157** and a plug portion **160** filling a remaining portion of the second contact holes **158** may be formed. Thus, the first contact plug **60**, including the lower layer **60***a*, the first metal-semiconductor compound layer **60***b*, and the upper layer **60***c* formed in each of the first contact holes **157**, and the second contact plugs **160***s* and **160**, including the second metal-semiconductor compound layer **160***s* and the plug portion **160** formed in each of the second contact holes **158**, may be formed.

(71) In an example, after the first metal-semiconductor compound layer **60***b* and the second metal-semiconductor compound layer **160***s* may be simultaneously formed on the lower layer **60***a* in the first contact holes **157** and the second sources/drains SD**2** exposed by the second contact hole **158**, respectively, a conductive material layer may be formed to fill remaining portions of the first contact holes **157** and a remaining portion of the second contact hole **158** and to cover the interconnection structures BS, the insulating fences **155**, and the upper insulating layer **128***c*. A separation insulating layer **65** may be formed to penetrate through the conductive material layer and to define the first and second pads **60***p* and **160***p*. For example, the separation insulating layer **65** may separate the conductive material layer into the first and second pads **60***p* and **160***p*. The separation insulating layer **65** may be formed of silicon nitride. The conductive material layer filling the remaining portions of the first contact holes **157** and the conductive material layer filling the remaining portions of the second contact holes **158** may be defined as the upper layer **60***c* and the plug portion **160**, respectively.

(72) According to the above-described method of fabricating a semiconductor device, the first contact plugs (**60** of FIGS. **13**A and **13**B) may be stably and reliably formed. For example, the insulating layer (**135** of FIGS. **5**A and **5**B) may be patterned using the first and second masks (**138**

and **141** of FIGS. **5**A and **5**B) formed of different materials. To remove the first and second mask structures **143***a* and **143***b* remaining after the insulating layer **135** is patterned to form the insulating patterns **135***a* in FIGS. **6**A and **6**B, **7**A to **9**B, as illustrated in FIGS. **7**A to **9**B, the sacrificial layer **148** may be formed, the second mask pattern in the first region MA may be removed by a dry etching process, a hydrogen plasma treatment process **151** may be performed to form the anti-oxidation layer (**141***p* of FIG. **9**B) on a surface of the second mask layer **141***b*′ in the second region PA, and the second mask layer **141***b*′, on which the anti-oxidation layer (**141***p* of FIG. **9**B) is formed, may be removed by the wet etching process. According to such a method, the insulating patterns **135***a* may be formed without damage or defects. Accordingly, the insulating fences (**155** of FIG. **12**B) formed between the insulating patterns **135***a* may be formed without deformation or defects and the insulating fences (**155** of FIG. **12**B) may be stably formed. Therefore, the first contact plugs (**60** of FIGS. **13**A and **13**B) formed between the insulating fences (**155** in FIG. **12**B) may be formed without defects. As a result, the first contact plugs (**60** of FIGS. **13**A and **13**B) may be stably and reliably formed.
(73) A semiconductor device **1**, fabricated by the above-described method of fabricating a semiconductor device, may be provided. The semiconductor device **1** may include the lower structure LS, the interconnection structures BS, the peripheral device structure TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129**, and the insulating fences **155**, the first contact plugs **60**, the second contact plugs **160***s* and **160**, the first and second pads **60***p* and **160***p*, and the separation insulating layer **65** described above.
(74) Hereinafter, various modified examples of the method of fabricating a semiconductor device according to an example embodiment will be described. Various modified examples to be described below will be mainly described with respect to modified or replaced components.
(75) A modified example of the method of fabricating a semiconductor device according to an example embodiment will be described with reference to FIGS. **14** and **15**A to **20** along with FIGS. **1**B and **2**. FIG. **14** is a process flowchart illustrating a modified example of the method for fabricating a semiconductor device according to an example embodiment, and FIGS. **15**A to **20** are cross-sectional views illustrating a modified example of the method for fabricating a semiconductor device according to an example embodiment. In FIGS. **15**A to **20**, FIGS. **15**A, **16**A, **17**A, and **18**A are cross-sectional views illustrating regions taken along lines I-I′ and II-II′ of FIG. **2**, and FIGS. **15**B, **16**B and **18**A, **17**B, **18**B, **19**, and **20** are cross-sectional views illustrating regions taken along lines and IV-IV′ of FIG. **2**.
(76) Referring to FIGS. **2**, **14**, **15**A, and **15**B, after operation S**30** of forming the insulating patterns **135***a* as described in FIGS. **1**A, **6**A, and **6**B, operation S**35** may be performed to form a protective layer **246**. The protective layer **246** may be formed to cover a side surface of at least the first mask pattern **138***a*. The protective layer **246** may be a silicon oxide layer. For example, the protective layer **246** may be a silicon oxide layer formed by an atomic layer deposition (ALD) process.
(77) In an example, the protective layer **246** may be formed to a thickness between about 20 angstroms and about 30 angstroms. Terms such as “about” or “approximately” may reflect amounts, sizes, orientations, or layouts that vary only in a small relative manner, and/or in a way that does not significantly alter the operation, functionality, or structure of certain elements. For example, a range from “about 0.1 to about 1” may encompass a range such as a 0%-5% deviation around 0.1 and a 0% to 5% deviation around 1, especially if such deviation maintains the same effect as the listed range.
(78) In an example, the protective layer **246** may conformally cover internal walls of the openings **145** and exposed surfaces of the first and second mask structures **143***a* and **143***b*.
(79) In an embodiment, the protective layer **246** may cover exposed surfaces of the first and second mask structures **143***a* and **143***b* without covering a portion of internal walls of the openings **145**.
(80) In an embodiment, grooves may be formed in upper regions of the cell gate capping layers **18** by the openings **145**, and the protective layer **246** may include portions **246***p* filling the grooves.

The portions **246***p* filling the grooves will be referred to as "protective patterns."
(81) In operation S**40**, a sacrificial layer **148** may be formed to fill a space between the insulating patterns **135***a*. The sacrificial layer **148** may be formed on the protective layer **246**.
(82) Then, a semiconductor device may be fabricated using a method substantially the same as or similar to the method as described with reference to FIGS. **1**B and **8**A to **13**B. Hereinafter, the method described with reference to FIGS. **1**B and **8**A to **13**B will be supplementarily described.
(83) Referring to FIGS. **1**B, **2**, **16**A, and **16**B, in operation S**55** of removing a second mask pattern **141***a* by a dry etching process in the process chamber as described with reference to FIGS. **1**A, **7**A, and **7**B, a first etching process may be performed to remove the protective layer (**246** of FIGS. **15**A and **15**B) disposed on upper surfaces of at least the second mask pattern **141***a* and the second mask layer **141***b*, instead of a native oxide layer which may be formed on a surface of the second mask pattern **141***a*. In the protective layer (**246** of FIGS. **15**A and **15**B), the remaining protective layer **246**′ may cover at least a portion of a side surface of the first mask pattern **138***a*.
(84) Referring to FIGS. **1**B, **2**, **17**A, and **17**B, when the second mask pattern **141***a* is formed of polysilicon, the second mask pattern **141***a* may be removed by a plasma etching process using a Cl.sub.2-based source gas. In this case, the second mask layer **141***b*′ having a second thickness T**2** as described with reference to FIG. **8**B may remain in the second region PA.
(85) Referring to FIGS. **1**B, **2**, **18**A, and **18**B, in operation S**60**, similarly to the description provided with reference to FIGS. **9**A and **9**B, a hydrogen plasma treatment process **151** may be performed to form the anti-oxidation layer **141***p* on a surface of the second mask layer **141***b*′ in the second region PA.
(86) Referring to FIGS. **1**B, **2**, and **19**, in operation S**70**, similarly to the description provided with reference to FIG. **10**, the second mask layer (**141***b*′ in FIG. **18**B), on which the anti-oxidation layer (**141***p* in FIG. **18**B) is formed, may be removed by a wet etching process.
(87) Referring to FIGS. **1**B, **2**, and **20**, a method substantially the same as the method as described with reference to FIGS. **11**A to **13**B may then be performed. For example, operation S**75** of removing the sacrificial layer **148** to form the fence holes **145** as described with reference to FIGS. **11**A and **11**B, operation S**80** of forming the insulating fences **155** in the fence holes **145** as described with reference to FIGS. **12**A and **12**B, operation S**85** of forming the first contact holes **157** as described with reference to FIGS. **13**A and **13**B, and operation S**90** of forming the first contact plugs **60** in the first contact holes **157** may be performed.
(88) In an embodiment, the second contact plugs **160***s* and **160**, the first and second pads **60***p* and **160***p*, and the separation insulating layer **65** as described with reference to FIGS. **13**A and **13**B may be formed.
(89) In an embodiment, the protective patterns **246***p* as described with reference to FIG. **15**B may remain below the insulating fences **155**. Accordingly, the protective patterns **246***p* may be formed below the insulating fences **155** and may be formed in the gate capping layers **18**.
(90) Accordingly, the semiconductor device **1**, fabricated by the method of fabricating a semiconductor device as described with reference to FIGS. **1**B, **2**, **14**, and **15**A to **20**, may be provided.
(91) Next, a modified example of the method of fabricating a semiconductor device according to an example embodiment will be described with reference to FIGS. **21** and **22**. FIGS. **21** and **22** are cross-sectional views illustrating regions taken along lines and IV-IV′ of FIG. **2**.
(92) Referring to FIG. **21** together with FIG. **2**, the protective layer described in FIGS. **15**A and **15**B (**246** in FIGS. **15**A and **15**B) may be replaced with a protective layer **346** conformally formed after forming the sacrificial layers (**148** of FIGS. **7**A and **7**B) as described in FIGS. **7**A and **7**B. The protective layer **346** may be formed of substantially the same material and have the same thickness as the protective layer (**246** of FIGS. **15**A and **15**B) as described with reference to FIGS. **15**A and **15**B.
(93) Referring to FIG. **22** together with FIG. **2**, similarly to the description provided with reference

to FIGS. **16**A and **16**B, in operation S**55** of removing the second mask pattern **141***a* by the dry etching process in the process chamber described with reference to FIGS. **7**A and **7**B, a first etching process may be performed to remove the protective layer (**346** of FIG. **21**) disposed on upper surfaces of at least the second mask pattern **141***a* and the second mask layer **141***b*, instead of a native oxide layer which may be formed on the surface of the second mask pattern **141***a*. In the protective layer **346** of FIG. **21**, a remaining passivation layer **346**′ may cover at least a portion of a side surface of the first mask pattern **138***a*. Then, the same method as described with reference to FIGS. **17**A to **20** may be performed.

(94) Accordingly, the semiconductor device **1** fabricated by the method of fabricating a semiconductor device, including the method in FIGS. **21** and **22**, may be provided.

(95) In modified embodiments, the protective layer (**246** of FIGS. **15**A and **15**B, or **346** of FIG. **21**) may protect a side surface of the first mask pattern **138***a* to prevent deformation of the first mask pattern **138***a*. Accordingly, the deformation of the first mask pattern **138***a* may be prevented, and the insulating patterns **135***a* may be formed without damage or defects. Accordingly, the insulating fences **155** formed between the insulating patterns **135***a* may be stably formed without deformation or defects, and the first contact plugs **60** formed between the insulating fences **155** may be formed without defects. As a result, the first contact plugs **60** may be stably and reliably formed.

(96) As described above, the method of fabricating a semiconductor device according to example embodiments may include forming the insulating layer **135** in the first region MA and the peripheral structure TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129** in the second region PA, sequentially forming the first mask layer **138** and the second mask layer **141** on the insulating layer **135** and the peripheral device structure TR**2**, **130**, **128***a*, **128***b*, **128***c*, and **129**, patterning the first and second mask layers **138** and **141** to form the first mask pattern **138***a* and the second mask pattern **141***a* sequentially stacked in the first region MA as well as to allow the first and second mask layers **138***b* and **141***b* to remain in the second region PA, etching the insulating layer **134** by an etching process using the mask structure **143***a* and **143***b*, including the first and second mask patterns **138***a* and **141***a* and the first and second mask layers **138***b* and **141***b*, as an etching mask to form the insulating pattern **135***a* in the opening **145**, removing the second mask pattern **141***a* in the first region MA by performing a dry etching process, forming an anti-oxidation layer **141***p* on the surface of the second mask layer **141***b*′ in the second region PA after removing the second mask pattern **141***a* in the first region MA, and removing the second mask layer **141***b*′ having a surface, on which the anti-oxidation layer **141***p* is formed, by performing the wet etching process. Then, the sacrificial layer **148** may be removed to form the fence holes **145**, insulating fences **155** may be formed in the fence holes **145**, and the first mask pattern **138***a* and the insulating pattern **135***a* may be removed to form a contact hole **157**, and the contact plug **60** may be formed in the contact hole **157**.

(97) As set forth above, according to embodiments, a method of fabricating a semiconductor device, including a method of stably patterning material layer using at least two mask layers, may be provided. A contact plug may be reliably formed without defects using the method of stably patterning the material layer.

(98) While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concept as defined by the appended claims.

Claims

1. A method of fabricating a semiconductor device, the method comprising: forming a plurality of bitline structures on a first region of a substrate; forming a peripheral device structure on a second region, adjacent to the first region, of the substrate; forming an insulating layer in spaces between two adjacent bitline structures of the plurality of bitline structures; forming a first mask layer and a

second mask layer sequentially on the insulating layer, the plurality of bitline structures, and the peripheral device structure; patterning the first and second mask layers to form a first mask structure including a first mask pattern and a second mask pattern sequentially stacked on the first region, and a second mask structure including the first and second mask layers remaining, after the patterning of the first and second mask layers, on the second region; etching the insulating layer by an etching process using the first and second mask structures as an etching mask, to form a plurality of insulating patterns in spaces between two adjacent bitline structures of the plurality of bitline structures; forming a sacrificial layer to fill spaces between two adjacent insulating patterns of the plurality of insulating patterns on the first region; removing the second mask pattern on the first region by performing a dry etching process; forming an anti-oxidation layer on a surface of the second mask layer on the second region after removing the second mask pattern on the first region; removing the second mask layer, having the surface on which the anti-oxidation layer is formed, by performing a wet etching process; removing the sacrificial layer to form a plurality of fence holes after selectively removing the second mask layer on which the anti-oxidation layer is formed; forming a plurality of insulating fences in the plurality of fence holes, respectively; removing the first mask pattern and the plurality of insulating patterns to form a plurality of contact holes; and forming a plurality of contact plugs in the plurality of contact holes, respectively.
2. The method of claim 1, wherein: the second mask layer is formed of polysilicon; and the forming of the anti-oxidation layer on the surface of the second mask layer on the second region comprises performing a hydrogen plasma treatment process on the surface of the second mask layer.
3. The method of claim 2, wherein the anti-oxidation layer is formed by terminating dangling bonds of the polysilicon of the second mask layer with hydrogen elements supplied by the hydrogen plasma treatment process.
4. The method of claim 1, wherein: the plurality of insulating patterns is formed of silicon oxide; and the plurality of insulating fences are formed of silicon nitride.
5. The method of claim 1, wherein a thickness of the second mask pattern on the first region is smaller than a thickness of the second mask layer remaining on the second region after the forming of the plurality of insulating patterns.
6. The method of claim 1, wherein the sacrificial layer, filling the spaces between the two adjacent insulating patterns of the plurality of insulating patterns, exposes at least a portion of a side surface of the second mask pattern.
7. The method of claim 1, wherein a height of the second mask layer on the second region is reduced during a time when the second mask pattern on the first region is removed by performing the dry etching process.
8. The method of claim 1, wherein the first mask layer on the second region is removed during a time when the plurality of contact holes are formed or after the plurality of contact holes are formed.
9. The method of claim 1, further comprising: forming a liner layer to cover a side surface of the first mask pattern before forming the sacrificial layer.
10. The method of claim 9, wherein: the liner layer covers internal walls of the spaces between the two adjacent insulating patterns of the plurality of insulating patterns; and at least a portion of the liner layer remains in a lower portion of each insulating fence of the plurality of insulating fences after forming the plurality of contact plugs.
11. The method of claim 1, further comprising: forming a liner layer to cover a side surface of at least the first mask pattern after forming the sacrificial layer.
12. A method of fabricating a semiconductor device, the method comprising: forming an insulating layer on a first region of a substrate and a peripheral structure on a second region of the substrate; forming a first mask layer and a second mask layer sequentially on the insulating layer and the peripheral structure; patterning the first and second mask layers to form a first mask structure

including a first mask pattern and a second mask pattern sequentially stacked on the first region, and a second mask structure including the first and second mask layers remaining, after the patterning of the first and second mask layers, on the second region; etching the insulating layer by an etching process using the first and second mask structures as an etching mask, to form a plurality of insulating patterns separated apart from each other; forming a sacrificial layer in spaces between two adjacent insulating patterns of the plurality of insulating patterns, on the first region; removing the second mask pattern on the first region by performing a dry etching process; forming an anti-oxidation layer on a surface of the second mask layer on the second region after removing the second mask pattern on the first region; and removing the second mask layer, having the surface on which the anti-oxidation layer is formed, by performing a wet etching process.

13. The method of claim 12, further comprising: removing the sacrificial layer to form a plurality of fence holes after selectively removing the second mask layer on which the anti-oxidation layer is formed; forming a plurality of insulating fences in the plurality of fence holes, respectively; removing the first mask pattern and the plurality of insulating patterns to form a plurality of contact holes; and forming a plurality of contact plugs in the plurality of contact holes, respectively.

14. The method of claim 12, further comprising: loading the substrate into a process chamber after forming the sacrificial layer, wherein the removing the second mask pattern on the first region by performing the dry etching process is performed in the process chamber, and wherein the forming the anti-oxidation layer on the surface of the second mask layer on the second region after removing the second mask pattern on the first region is performed in the process chamber; and unloading the substrate, after forming the anti-oxidation layer on the surface of the second mask layer, from the process chamber.

15. The method of claim 14, wherein: the second mask layer is formed of polysilicon; and the forming the anti-oxidation layer on the surface of the second mask layer on the second region comprises performing a hydrogen plasma treatment process on the surface of the second mask layer in the process chamber.

16. The method of claim 15, wherein: wherein the anti-oxidation layer is formed by terminating dangling bonds of silicon elements of the polysilicon of the second mask layer with hydrogen elements supplied by the hydrogen plasma treatment process.

17. The method of claim 13, wherein: the plurality of insulating patterns are formed of silicon oxide; and the plurality of insulating fences are formed of silicon nitride.

18. A method of fabricating a semiconductor device, the method comprising: forming a cell transistor at a first region of a substrate; forming a plurality of bitline structures and a peripheral device structure on the substrate, wherein the plurality of bitline structures are formed on the first region and the peripheral device structure is formed on a second region, adjacent to the first region, of the substrate; forming an insulating layer between spaces between two adjacent bitline structures of the plurality of bitline structures; forming a first mask layer and a second mask layer sequentially on the insulating layer, the plurality of bitline structures, and the peripheral device structure; patterning the first and second mask layers to form a first mask structure including a first mask pattern and a second mask pattern sequentially stacked on the first region, and a second mask structure including the first and second mask layers remaining, after the patterning of the first and second mask layers, on the second region; etching the insulating layer by an etching process, using the first and second mask structures as an etching mask, to form a plurality of insulating patterns, wherein each of the plurality of insulating patterns is disposed between corresponding two adjacent bitline structures of the plurality of bitline structures; forming a sacrificial layer to fill spaces between two adjacent insulating patterns of the plurality of insulating patterns, on the first region; removing the second mask pattern by performing a dry etching process; forming an anti-oxidation layer on a surface of the second mask layer on the second region after removing the second mask pattern; and removing the second mask layer, having the surface on which the anti-oxidation layer is formed, by performing a wet etching process.

19. The method of claim 18, further comprising: removing the sacrificial layer to form a plurality of fence holes after selectively removing the second mask layer on which the anti-oxidation layer is formed; forming a plurality of insulating fences in the plurality of fence holes, respectively; removing the first mask pattern and the plurality of insulating patterns to form a plurality of contact holes; and forming a plurality of contact plugs in the plurality of contact holes, respectively, wherein each bitline structure of the plurality of bitline structures comprises a bitline, and wherein the cell transistor comprises a wordline.

20. The method of claim 18, further comprising: loading the substrate with the sacrificial layer into a process chamber after forming the sacrificial layer to fill the spaces between the two adjacent insulating patterns of the plurality of insulating patterns on the first region, wherein the removing the second mask pattern by performing the dry etching process is performed in the process chamber, and wherein the forming the anti-oxidation layer on the surface of the second mask layer on the second region after removing the second mask pattern is performed in the process chamber; and unloading the substrate with the anti-oxidation layer, after forming the anti-oxidation layer on the surface of the second mask layer, from the process chamber.